

Osmotic modulus of a PDDA/BSA coacervate layer from PEG osmotic compression

Heights at 80 h, 0–15% PEG (n = 3 per group) · nominal PEG osmotic pressures · 20% PEG reported in the SI only

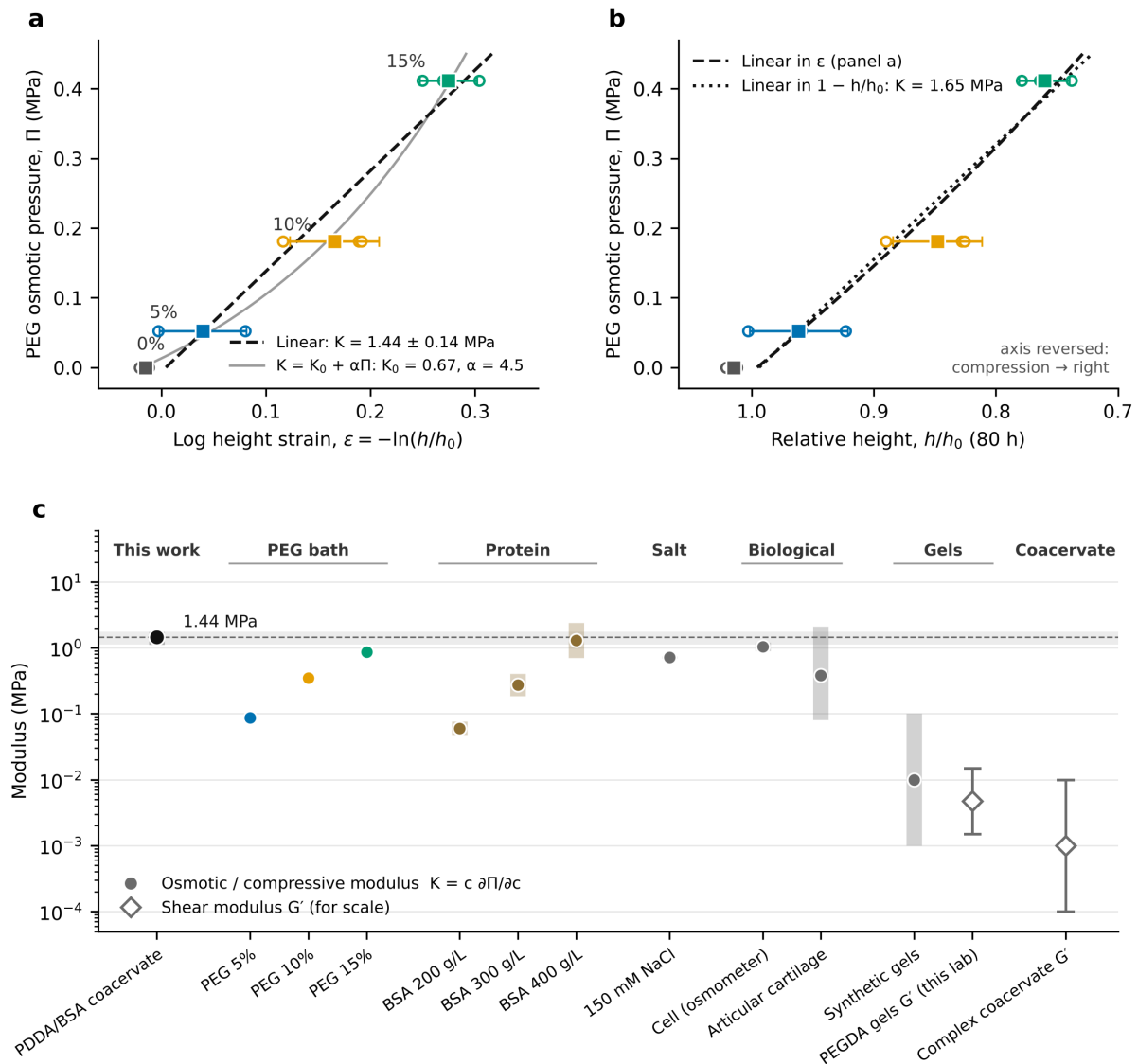


Figure 1 | The PDDA/BSA coacervate resists osmotic dehydration with a modulus of about 1.4 MPa, like a very crowded protein fluid. (a) PEG osmotic pressure Π against log height strain $\varepsilon = -\ln(h/h_0)$ of the coacervate layer after 80 h, for 0% (grey), 5% (blue), 10% (orange) and 15% (green) PEG. Open circles: individual samples. Squares: group mean $\pm$ SD (n = 3). Because the layer is laterally confined, $V/V_0 = h/h_0$ and the slope is the osmotic modulus $K = c\partial\Pi/\partial c$. Dashed line: linear fit to all 12 samples, strain regressed on Π , $K = 1.44 \pm 0.14$ MPa (95% CI 1.13–1.76). Grey curve: stiffening equation of state $K = K_0 + \alpha\Pi$ ($K_0 = 0.67$ MPa, $\alpha = 4.5$; $\Delta\text{AlCc} = -1.8$ against the linear fit, so the two are statistically indistinguishable). **(b)** The same data against plain relative height h/h_0 . The axis is reversed so compression runs to the right, as in (a). Dashed: the panel-a fit mapped onto h/h_0 . Dotted: linear fit in engineering strain $1 - h/h_0$, $K = 1.65 \pm 0.18$ MPa. **(c)** The coacervate modulus (black; bar = 95% CI, grey band carried across the panel) compared with benchmarks from soft matter, log scale. Filled circles: osmotic or confined-compression moduli. The PEG baths used here are computed from the same $\Pi(w)$ curve, in the colours of (a). BSA is a Carnahan–Starling hard-sphere estimate ($M = 66.4$ kDa; bar spans $v_{\text{eff}} = 1.2\text{--}1.5$ mL g $^{-1}$). 150 mM NaCl is ideal van ’t Hoff with $K = \Pi$. The cell is treated as a Boyle–van ’t Hoff osmometer at 290 mOsm with an osmotically inactive fraction of 0.2–0.4. Articular cartilage is the equilibrium aggregate modulus in confined compression (range over depth; dot = full thickness, Schinagl et al. 1997). Synthetic gels: typical osmotic moduli of swollen gels. Open diamonds: shear moduli, shown for scale only. PEGDA gels are this lab’s G' data (1.5–15 kPa). Complex coacervates are typical plateau G' values (Spruijt et al. 2013).

What the figure shows, and how the panels build the argument

Focus. Pressing a coacervate layer with PEG is an osmotic-stress experiment. The PEG sets the water activity, the layer gives up water until its own osmotic pressure matches, and the height change reports how much volume that took. The figure asks one question: **how hard is it to squeeze water out of this coacervate, and is that hard compared with other soft materials?** The answer is $K \approx 1.4$ MPa. That is about 1500× softer than liquid water, so the coacervate is not incompressible. It is nonetheless as stiff as a ~400 g/L protein solution or a living cell, and 2–4 orders of magnitude stiffer than gels or the coacervate's own shear response.

Logic arc through the panels

a • Measure the equation of state. With lateral confinement, height is volume, and $-\ln(h/h_0)$ is the natural compression variable. Its slope against Π is the thermodynamic osmotic modulus with no geometric correction. Over 0–15% PEG the response is close to linear ($K = 1.44$ MPa). The step modulus doubles between 10% and 15%. A stiffening equation of state fits equally well, so the data are honestly described as a small-strain modulus over a 1.3-fold concentration window, possibly stiffening toward its end.

b • Show the result is not an artefact of the log transform. Plotted against plain h/h_0 , the same points give 1.65 MPa with an engineering-strain fit. The two strain measures differ by at most 0.04 here, so the choice of axis changes K by about 15%, not its order of magnitude. Panel b is also the form most readers will check against the raw images.

c • Calibrate the number. A modulus means little without a scale. On one log axis the coacervate sits with crowded liquids: about 400 g/L BSA, a cell under osmotic load, physiological saline, and the stiff end of cartilage. It sits far above swollen gels and above any shear modulus a coacervate liquid shows. The contrast with shear is the physical point: a coacervate flows, yet it holds onto its water as tightly as a dense protein phase.

Take-home. $K_{\text{osm}} \approx 1.4$ MPa (1.1–1.8, 95% CI) from 0–15% PEG. The coacervate dehydrates reversibly and smoothly under osmotic stress, with a compressibility typical of a very concentrated protein or polyelectrolyte fluid. 20% PEG gives a collapse that this equation of state does not predict, together with wall climbing. It is reported in the SI but not used for the modulus.

Supporting notes: method, fits, linearity, benchmarks and caveats

Method

- The layer can only change height, so $V/V_0 = h/h_0$. At equilibrium the PEG pressure Π equals the coacervate's own osmotic pressure, so Π against strain is the coacervate equation of state.
- The osmotic modulus is $K = c \partial \Pi / \partial c = d\Pi / d\varepsilon$, with $\varepsilon = -\ln(h/h_0)$.
- Heights are the 80 h values of each sample ($n = 3$ per group, 0–15% PEG). They were digitised from the original per-sample panels. Where markers overlapped, the hidden value was solved from the plotted group mean.
- Strain is regressed on Π and the slope is inverted. Π is the variable you set; ε is the one that scatters. Regressing Π on ε biases K when ε is noisy.
- **20% PEG is not used.** The layer kept climbing the wall, so height no longer tracks volume. Sample 2 is also inconsistent ($h/h_0 = 0.88$, less compressed than any 15% sample). All of the 20% data is shown in the SI figure.

Results

Model (0–15% PEG, $n = 12$)	K (MPa)	95% CI
Linear in log strain (panel a)	1.44 ± 0.14	1.13–1.76
Linear in engineering strain $1 - h/h_0$ (panel b)	1.65 ± 0.18	1.26–2.04
Stiffening, $K = K_0 + \alpha \Pi$	$K_0 = 0.67 \pm 0.25$, $\alpha = 4.5 \pm 2.0$	K at 15% ≈ 2.5

The step moduli between group means are 0.95, 1.03 and 2.12 MPa. The stiffening model fits slightly better ($\Delta\text{AICc} = -1.8$), but a difference under 2 is not decisive with $n = 12$.

Is Π linear in $\ln(h/h_0)$ over a wide range, and is that common?

The range is narrower than it looks. The 15% layer is at $h/h_0 = 0.76$, which is a 1.3-fold increase in concentration. Over a window that small, almost any smooth equation of state looks linear. Write the modulus as $K(\epsilon) = K_0 \cdot \exp(\beta\epsilon)$, where $\beta = d \ln K / d \ln c$. The secant over the window then exceeds K_0 by only $(e^{\beta\epsilon} - 1)/\beta\epsilon$. For a semidilute polymer in good solvent ($\Pi \propto c^{9/4}$, $\beta \approx 2.3$) the secant at $\epsilon = 0.29$ is only about 40% above K_0 . With $n = 3$ and an SD of about 0.04 in ϵ , that much curvature is hard to see in a plot.

Your data hints at the curvature. The step modulus doubles between 10% and 15%. The stiffening fit gives $\alpha = 4.5$, which means K at 15% is about 2.5 MPa.

Over wide ranges, other forms are linear instead. Osmotic stress data spanning decades are usually linear in $\log \Pi$ against $\log c$ (a power law: polymers, gels, proteins), or in $\log \Pi$ against spacing (exponential hydration forces between DNA or lipid layers). A plot linear in Π against $\ln c$ would mean a constant K , which is the small-strain limit, not a general law.

Why the coacervate has a finite K_0 . A coacervate in equilibrium with its dilute phase starts at $\Pi = 0$, not at some prestressed state. So K_0 is finite and the response begins linear. Report K as the small-strain modulus over a stated window: 1.44 MPa for 0–0.41 MPa and $\epsilon \leq 0.3$.

Which axis for the height?

Keep Π on the y-axis and strain on the x-axis. The slope is then the modulus, in the same orientation as a stress-strain curve. Do the fit the other way round (strain on Π), because Π is the controlled variable. Panel b shows the same data against plain h/h_0 , with the axis reversed so compression still runs to the right. At these strains the log and linear measures differ by at most 0.04 in strain. The engineering-strain modulus is 1.65 MPa against 1.44 MPa.

How stiff is ~1.4 MPa? (panel c)

- **Osmotic moduli of crowded liquids:** the coacervate matches about 400 g/L BSA (hard-sphere estimate 0.7–2.4 MPa). It also matches the osmotic stiffness of a cell treated as an osmometer ($\Pi/(1 - b)$ at 290 mOsm $\approx$ 0.9–1.2 MPa) and of 150 mM NaCl ($K = \Pi \approx$ 0.7 MPa). It is stiffer than the 15% PEG bath compressing it (0.87 MPa).
- **Confined-compression moduli:** it sits at the top of the range for articular cartilage (0.08–2.1 MPa, depth dependent; 0.38 MPa for full thickness).
- **Gels:** it is 1–3 orders of magnitude stiffer than the osmotic moduli of typical swollen synthetic gels (about 1–100 kPa) and the shear moduli of your PEGDA gels (1.5–15 kPa).
- **Coacervate shear:** complex coacervates are viscoelastic liquids with plateau G' of roughly 10^2 – 10^4 Pa. They flow in shear yet resist water removal at the MPa scale.

So the coacervate is not incompressible. Liquid water has a bulk modulus of 2.2 GPa, about 1500 $\times$ higher. It is osmotically stiff in the sense of a very crowded protein or polyelectrolyte fluid.

Caveats

- The PEG pressures are nominal. PEG entering the coacervate, or salt redistribution, would lower the effective Π and therefore K .
- The 15% layer rebounds by about 0.02 in h/h_0 between 32 h and 80 h. Using the minimum height lowers its secant modulus by about 8%.
- The BSA values are a Carnahan-Starling estimate ($M = 66.4$ kDa, $v_{\text{eff}} = 1.2$ – 1.5 mL/g), not measured data. The synthetic-gel and coacervate- G' bars are typical literature ranges and are meant for scale only. Cartilage values: Schinagl et al., J. Orthop. Res. 1997. Coacervate rheology: Spruijt et al., Macromolecules 2013.
- The heights are digitised from images. The reading error is about 0.005 in h/h_0 , and some hidden points were solved from the group means. Replace them with the measured values before publishing.

Supplementary Information

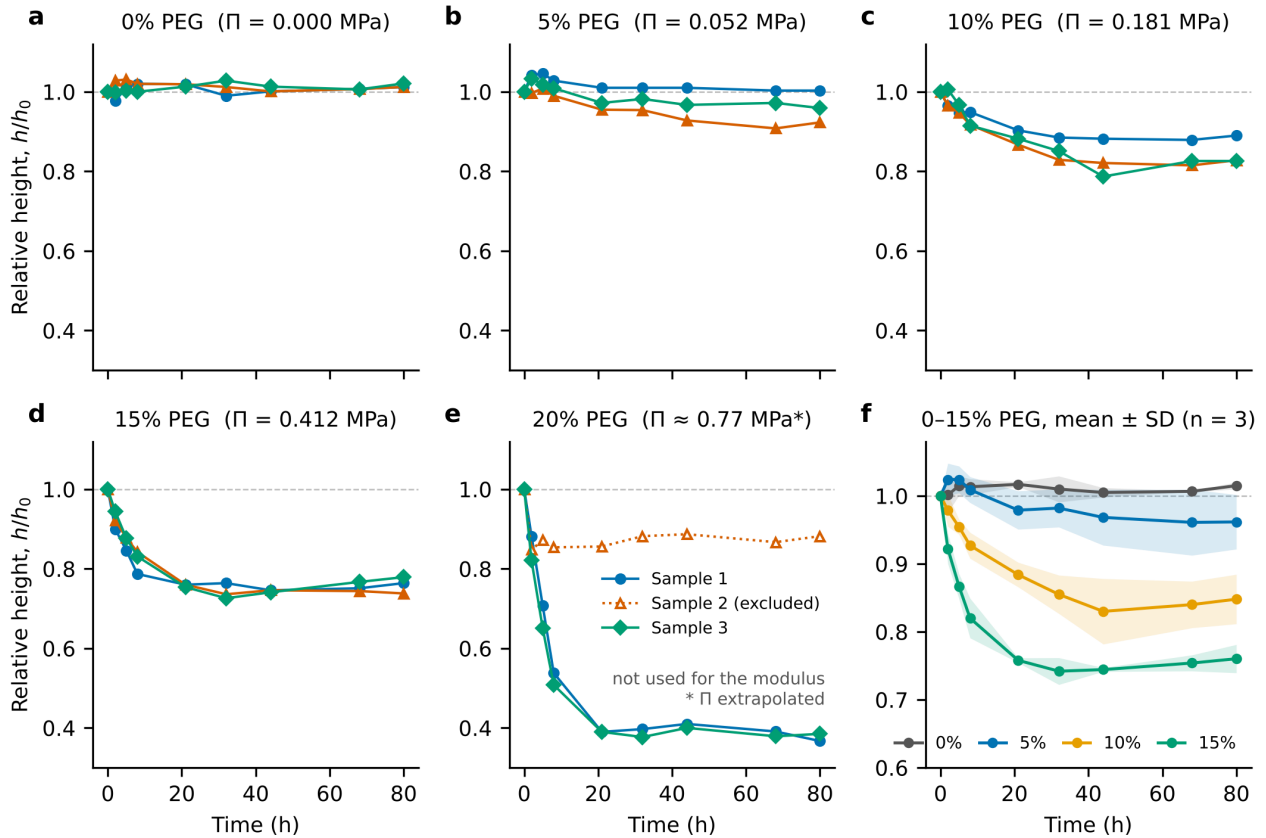


Figure S1 | Relative height h/h_0 of every coacervate layer over 80 h of PEG osmotic stress. (a-e) Individual samples at 0, 5, 10, 15 and 20% PEG (sample 1 blue circles, sample 2 orange triangles, sample 3 green diamonds). The panel titles give the nominal PEG osmotic pressure. For 20% it is extrapolated (*) from a Rand-type fit $\log_{10}\Pi = a + b \cdot w^{0.21}$ through the 5–15% values. At 20% PEG sample 2 (open triangles, dotted) stalls at $h/h_0 \approx 0.88$, less compressed than any 15% sample, and is excluded. Samples 1 and 3 collapse to $h/h_0 \approx 0.38$ while the layer climbs the wall, so height no longer reports volume. The 20% group is therefore not used for the modulus. **(f)** Mean $\pm$ SD ($n = 3$) for 0–15% PEG. The 80 h values are the inputs to Figure 1. Data were digitised from the original per-sample plots (reading error ≈ 0.005). Values hidden by overlapping markers were solved from the plotted group means.

Table S1 | Digitised h/h_0 for every sample and time point

PEG	Sample	0 h	2 h	5 h	8 h	21 h	32 h	44 h	68 h	80 h
0%	1	1.000	0.977	1.011	1.020	1.019	0.990	1.001	1.007	1.012
0%	2	1.000	1.028	1.031	1.020	1.019	1.012	1.001	1.007	1.012
0%	3	1.000	0.999	1.003	1.000	1.013	1.028	1.013	1.006	1.021
5%	1	1.000	1.042	1.046	1.028	1.010	1.010	1.010	1.003	1.003
5%	2	1.000	0.997	1.008	0.990	0.955	0.954	0.928	0.908	0.923
5%	3	1.000	1.033	1.018	1.009	0.972	0.982	0.967	0.972	0.959
10%	1	1.000	0.965	0.948	0.949	0.903	0.885	0.882	0.879	0.890
10%	2	1.000	0.965	0.948	0.917	0.867	0.829	0.821	0.815	0.828
10%	3	1.000	1.006	0.967	0.915	0.882	0.851	0.787	0.826	0.826
15%	1	1.000	0.900	0.846	0.787	0.760	0.764	0.746	0.751	0.764
15%	2	1.000	0.922	0.878	0.842	0.760	0.736	0.746	0.744	0.738
15%	3	1.000	0.944	0.877	0.831	0.754	0.726	0.741	0.767	0.779
20%	1	1.000	0.882	0.708	0.538	0.390	0.397	0.410	0.391	0.367
20%	2 (excl.)	1.000	0.848	0.872	0.854	0.856	0.882	0.887	0.867	0.882
20%	3	1.000	0.821	0.651	0.508	0.390	0.377	0.400	0.379	0.385